SEBASTIAN ALEXANDER ESCURRA CARMELO

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PROFILE

Second-year Mathematics student at the University of Montpellier, previously in Physics at UNI (Lima, Peru), moving towards quantitative finance. Built two projects in Monte Carlo option pricing and order book simulation. Trilingual (English, French, Spanish), with experience as a private math tutor. Seeking a research internship in quantitative finance, market design and applied mathematics.

EDUCATION

University of Montpellier – Bachelor's Degree (L2), Mathematics — Sep 2026 – Present
Montpellier, France

University of Montpellier – Bachelor's Degree (L1), Physics — Sep 2025 – May 2026
Montpellier, France
Ranked 8th out of 273 students
Ranked 1st out of 547 students in Outils Mathématiques 1 (20/20)

National University of Engineering (UNI) – Physics, Bachiller program — Aug 2019 – Feb 2025
Part-time while working: six of ten semesters taken, program not completed | Lima, Peru

WORK EXPERIENCE

Private Mathematics Tutor — Aug 2019 – Aug 2025
Lima, Peru

- Provided individual and small-group tutoring in mathematics at pre-university and university level.
- Supported students in calculus, linear algebra and other undergraduate mathematics topics.
- Adapted explanations and problem-solving approaches to students with different levels of preparation.

PROJECTS

Order Book & Matching Engine Simulator – Java — 2026

- Built an object-oriented order book simulator in Java (Order, Trade, OrderBook, MatchingEngine) with price-time priority matching, partial fills and order cancellation.
- Validated all inputs and wrote automated tests, including randomized checks that the book is never crossed and that traded quantity is conserved.

Monte Carlo Option Pricing – Python — 2026

- Priced a European call under the Black-Scholes model by simulating the terminal stock price, with 95% confidence intervals, and compared it with the exact analytical price.
- Implemented antithetic variates and a control variate; combined, they reduced the variance of the estimate by about 21 times for the same computing budget.
- Verified the $1/\sqrt{n}$ convergence rate and checked that the confidence intervals contain the true price about 95% of the time.

SKILLS

Programming: Python (NumPy, pandas, matplotlib, SciPy), Java, Object-Oriented Programming
Quantitative: Probability, Statistics, Linear Algebra, Numerical Computing, Confidence Intervals
Finance: Option Pricing (Black-Scholes), Monte Carlo Simulation, Variance Reduction, Order Book Mechanics

LANGUAGES

Spanish (native) | English (C1) | French (C1)

INTERESTS

Personal Investing – Monthly investments of my own savings in an S&P 500 index ETF.
Chess & Poker – Probability, expected-value reasoning and decision-making under uncertainty.
Competitive Boxing – Fight Academy Boxing Club; amateur national champion, most recently in Paris (Mar 2026).